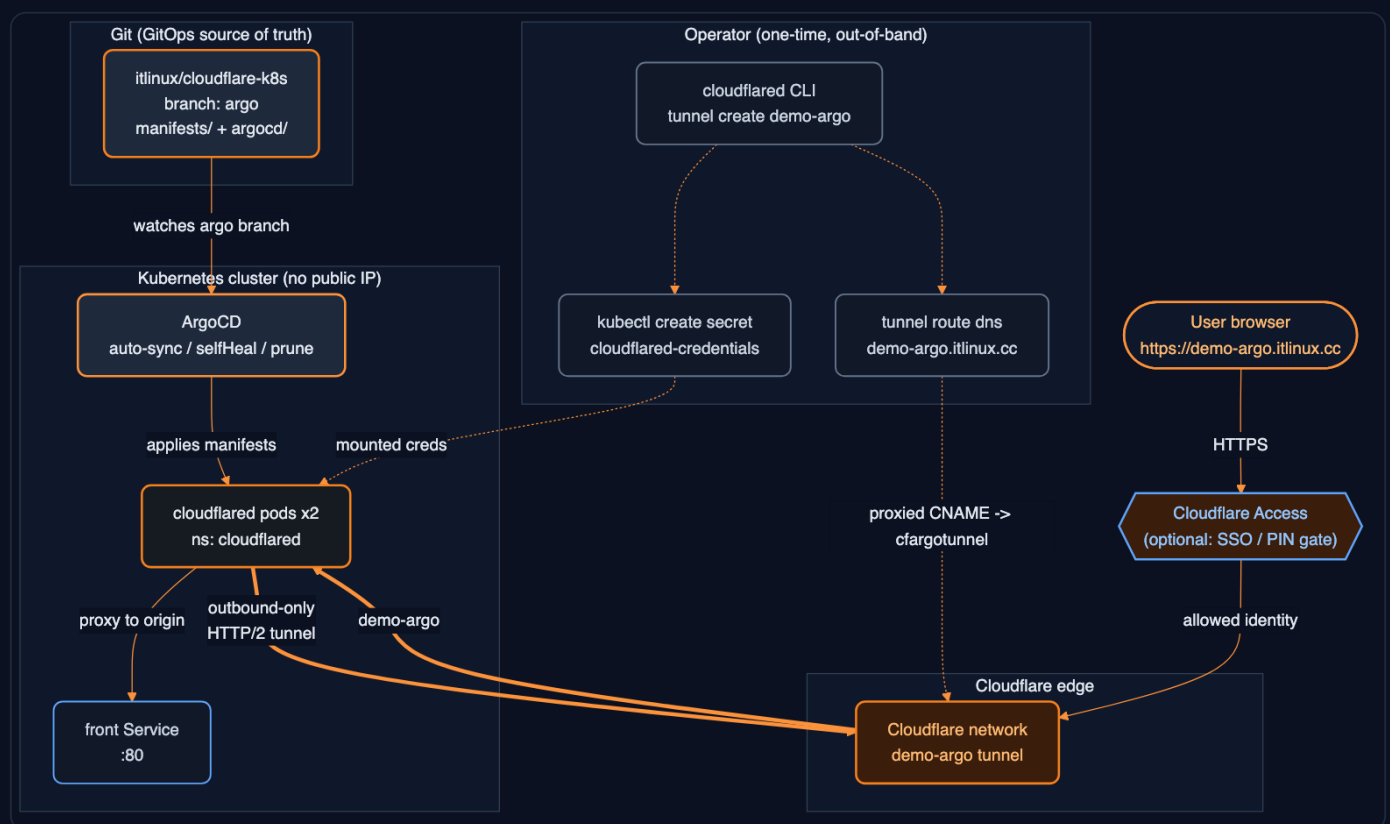


Cloudflare Tunnel on Kubernetes — ArgoCD GitOps (demo-argo)

Deploy a Cloudflare Tunnel connector (`cloudflared`) into a Kubernetes cluster with **ArgoCD**, so Services are reachable on a public hostname **without exposing any public IP / LoadBalancer**. The tunnel (`demo-argo`) is built manually; ArgoCD manages the in-cluster connector via GitOps.

Architecture

Internet → Cloudflare edge → demo-argo tunnel → cloudflared pods → Service
(DNS: demo-argo.itlinux.cc) (outbound-only) (in cluster) (front:80)



Flow, end to end:

1. **GitOps source** — the `argo` branch holds `manifests/` (cloudflared Deployment + ConfigMap) and `argocd/` (the Application CR).
2. **ArgoCD** watches the `argo` branch and applies the manifests into the `cloudflared` namespace (auto-sync, selfHeal, prune).
3. **Operator (one-time, out-of-band)** — builds the `demo-argo` tunnel with the `cloudflared` CLI, loads its `credentials.json` as a Secret, and creates the DNS route. None of this is in git.
4. **cloudflared pods** mount the creds Secret and open an **outbound-only HTTP/2 tunnel** to the Cloudflare edge — no inbound ports, no public IP.
5. **User traffic** hits `https://demo-argo.itlinux.cc` → Cloudflare edge → the `demo-argo` tunnel → a cloudflared pod → the in-cluster `front` Service.
6. **No inbound exposure** — `cloudflared` dials *out* to Cloudflare; nothing is opened on the cluster.
7. **ArgoCD** continuously reconciles the connector Deployment/ConfigMap to the `argo` branch.
8. **Tunnel credentials** live in a Kubernetes Secret created **out-of-band** — never in git.

Repository layout (`argo` branch)

```
manifests/
  cloudflared.yml      # Deployment (2 replicas, HA) + ConfigMap (tunnel: demo-argo)
  kustomization.yaml   # pins the cloudflared image tag (no :latest drift)
argocd/
  application.yaml     # ArgoCD Application CR – points at manifests/
  root-app.yaml        # App-of-Apps root – adopts everything in argocd/
terraform/             # optional IaC: DNS route + Access gate (hybrid)
README-argocd.md       # this document
.gitignore             # keeps credentials.json out of git
```

Prerequisites

- A running Kubernetes cluster with `kubectl` access (EKS, GKE, k3s, ...).
- **ArgoCD** installed in the cluster (namespace `argocd`).
- `cloudflared` CLI on your workstation.
- A domain managed in Cloudflare (this guide uses `demo-argo.itlinux.cc`).

Step 0 — Connect to your existing EKS cluster (AWS CLI)

This deployment runs on **Amazon EKS**. Wire up cluster access first; every `kubectl` step below then targets the EKS cluster.

Install the tooling

```
# macOS (Homebrew). eksctl only needed if you ever create/manage clusters.
brew install awscli kubectl helm

aws --version      # aws-cli/2.x
kubectl version --client
```

Authenticate to AWS

```
aws configure      # access key, secret, default region (e.g. us-west-2)
# ...or SSO:
aws sso login --profile <your-profile>
aws sts get-caller-identity  # confirm who you are
```

Point kubectl at the EKS cluster

```
aws eks list-clusters --region <region>
# Write/refresh the kubeconfig context for the cluster:
aws eks update-kubeconfig --name <cluster-name> --region <region>

kubectl config current-context      # arn:aws:eks:<region>:<acct>:cluster/<name>
kubectl get nodes                  # confirm the cluster is reachable
```

Install ArgoCD on the cluster (if not already present)

```
kubectl create namespace argocd
kubectl apply -n argocd \
```

```
-f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
kubectl -n argocd rollout status deploy/argocd-server
```

EKS networking note: EKS uses the AWS VPC CNI (pod IPs from the VPC, ~9001 MTU) — `cloudflared` works fine. The ConfigMap already sets `protocol: http2` as a safe default; if you swap to an overlay CNI (Calico/Cilium overlay) that reduces MTU, keep `http2` to avoid QUIC datagram flaps.

Step 1 — Build the tunnel + credentials Secret (one-time, NOT GitOps-managed)

The `demo-argo` tunnel is created manually. Its `credentials.json` becomes the `cloudflared-credentials` Secret that the Deployment mounts. The ConfigMap's `tunnel: demo-argo` must match this tunnel.

Where does `cloudflared` run to create the tunnel?

Creating the tunnel and running the connector are **two separate things**:

- **Creating the tunnel** is a one-time admin action. You run the `cloudflared` CLI **wherever you have it** — your **laptop/workstation** is fine. It does **not** have to run on the cluster or a server. The CLI just registers the tunnel with Cloudflare and writes the `credentials.json` locally.
- **Running the connector** is what actually serves traffic — that's the `cloudflared` **Pods in Kubernetes** (managed by ArgoCD). They consume the credentials you generated.

So: create on your laptop → load the creds into the cluster → ArgoCD runs the connector. You can even create the tunnel on one machine and never run a connector there at all.

Two ways to create the tunnel

Option 1 — CLI (from your laptop):

```
cloudflared login          # browser auth — pick the itlinux.cc zone
cloudflared tunnel create demo-argo # writes ~/.cloudflared/<TUNNEL-ID>.json

kubectl create namespace cloudflared

# Load the tunnel creds as the Secret (key name MUST be credentials.json)
TID=$(cloudflared tunnel list --output json \
  | python3 -c "import sys,json;print([t['id'] for t in json.load(sys.stdin) if t['name']=='demo-argo'][0])")
kubectl -n cloudflared create secret generic cloudflared-credentials \
  --from-file=credentials.json="$HOME/.cloudflared/${TID}.json"

# Route the public hostname to the tunnel (creates the proxied CNAME)
cloudflared tunnel route dns demo-argo demo-argo.itlinux.cc
```

Option 2 — Dashboard (no CLI needed):

Zero Trust → **Networks** → **Connectors** → **Create a connector** → **Cloudflared** → name it `demo-argo`. (The dashboard section formerly called **Tunnels** is now **Connectors** — `dash → Networks → Connectors`; the underlying resource is still a Cloudflare Tunnel and the CLI verbs are still `cloudflared tunnel ...`.) The dashboard shows an install/run command containing a **connector token** — for a dashboard-created (“remotely-managed”) connector you use that **token** instead of a `credentials.json` file:

```
# Dashboard-managed connector: store the token as the Secret instead of creds file
kubectl create namespace cloudflared
kubectl -n cloudflared create secret generic cloudflared-credentials \
  --from-literal=token='<CONNECTOR-TOKEN-from-dashboard>'
```

Then run the connector with `tunnel run --token` instead of a config file. If you go this route, change the Deployment args to: `["tunnel", "run", "--token", "${TUNNEL_TOKEN}"]` and inject the token via an env var from the Secret. Public hostnames + routes are configured in the dashboard's **Connectors** UI rather than the ConfigMap `ingress:` block.

Terminology (2026): the dashboard menu is **Networks** → **Connectors** (was *Tunnels*). The CLI is unchanged (`cloudflared tunnel create/run/route`), and the resource is still a “Cloudflare Tunnel.” This doc says **connector** for the dashboard/runtime component and **tunnel** for the CLI/resource.

This README's manifests use the **CLI / credentials-file** model (Option 1), which keeps the routing config in git (the ConfigMap `ingress`). The dashboard token model moves routing into the Cloudflare UI — pick one, don't mix.

Never commit `credentials.json` or the tunnel token. They are the tunnel secret. ArgoCD is configured to ignore this Secret so a sync/prune never deletes or manages it.

Step 2 — The Kubernetes manifest (`manifests/cloudflared.yml`)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudflared
  labels: { app: cloudflared }
spec:
  selector:
    matchLabels: { app: cloudflared }
  replicas: 2 # HA: 2 connectors share the same tunnel
  template:
    metadata:
      labels: { app: cloudflared }
    spec:
      containers:
        - name: cloudflared
          image: cloudflare/cloudflared:latest # tag pinned via kustomization.yaml
          args: ["tunnel", "--config", "/etc/cloudflared/config.yaml", "run"]
          livenessProbe:
            httpGet: { path: /ready, port: 2000 }
            failureThreshold: 1
            initialDelaySeconds: 10
            periodSeconds: 10
          resources:
            requests: { cpu: 50m, memory: 64Mi }
            limits: { cpu: 200m, memory: 128Mi }
          volumeMounts:
            - { name: config, mountPath: /etc/cloudflared, readOnly: true }
            - { name: creds, mountPath: /etc/cloudflared/creds, readOnly: true }
      volumes:
        - name: creds
          secret: { secretName: cloudflared-credentials }
        - name: config
          configMap:
            name: cloudflared
            items: [{ key: config.yaml, path: config.yaml }]
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: cloudflared
data:
  config.yaml: |
    tunnel: demo-argo
    protocol: http2 # EKS/overlay-CNI safe (avoids QUIC MTU flaps)
    metrics: 0.0.0.0:2000
    no-autoupdate: true
    credentials-file: /etc/cloudflared/creds/credentials.json
    ingress:
      - hostname: "demo-argo.itlinux.cc"
```

```
service: http://front.default.svc.cluster.local:80
- service: http_status:404
```

`manifests/kustomization.yaml` pins the image so syncs are reproducible:

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
- cloudflared.yaml
images:
- name: cloudflare/cloudflared
  newTag: "2026.6.1"
```

Step 3 — The ArgoCD Application (`argocd/application.yaml`)

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: cloudflared
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  project: default
  source:
    repoURL: https://github.com/itlinux/cloudflare-k8s.git
    targetRevision: argo          # the branch
    path: manifests              # kustomize dir
  destination:
    server: https://kubernetes.default.svc
    namespace: cloudflared
  syncPolicy:
    automated:
      prune: true                # remove resources deleted from git
      selfHeal: true             # revert manual drift back to git
    syncOptions:
    - CreateNamespace=true
  ignoreDifferences:             # never manage/prune the out-of-band Secret
  - group: ""
    kind: Secret
    name: cloudflared-credentials
    jsonPointers: ["/data"]
```

Step 4 — Register the app (GitOps, no ArgoCD CLI)

You do **not** need the `argocd` CLI. There are two pure-GitOps ways to register the Application so ArgoCD pulls the repo and reconciles automatically.

4a. One-time bootstrap (kubectl apply the Application CR)

A single declarative apply — after this, everything is git-driven:

```
kubectl apply -f argocd/application.yaml
```

That CR points at the `argo` branch / `manifests` path with `automated` sync, so ArgoCD immediately pulls and deploys, then keeps reconciling on every push. No CLI, no manual sync. This one apply is the only imperative step — and even it can be eliminated with App-of-Apps below.

4b. App-of-Apps — zero per-app registration (recommended for CI/CD)

Point ArgoCD at a **single root Application** once; from then on you add new apps just by committing YAML — ArgoCD discovers and deploys them. Nothing imperative per app, ever.

`argocd/root-app.yaml` (the only thing you ever register):

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: root
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/itlinux/cloudflare-k8s.git
    targetRevision: argo
    path: argocd          # ArgoCD watches this DIR for child Applications
  destination:
    server: https://kubernetes.default.svc
    namespace: argocd
  syncPolicy:
    automated: { prune: true, selfHeal: true }
```

Because `argocd/application.yaml` (the cloudflared app) lives in that `argocd/` dir, the root app picks it up automatically. To add another service later, drop a new `argocd/<name>.yaml` and push — ArgoCD deploys it with no extra commands.

4c. Connect the repo to ArgoCD (declarative, for private repos)

If the repo is private, register the credentials as a Secret (also GitOps-able) rather than `argocd repo add`:

```
apiVersion: v1
kind: Secret
metadata:
  name: cloudflare-k8s-repo
  namespace: argocd
  labels:
    argocd.argoproj.io/secret-type: repository
stringData:
  type: git
  url: https://github.com/itlinux/cloudflare-k8s.git
  password: <github-pat>      # or use SSH key fields for git@ URLs
  username: git
```

Public repo? Skip this — ArgoCD pulls anonymously.

Verify (kubectl only, no argocd CLI)

```
kubectl -n argocd get applications          # SYNCED / HEALTHY
kubectl -n cloudflared get pods             # cloudflared running
kubectl -n cloudflared logs -l app=cloudflared --tail=20  # "Registered tunnel connection"
curl -I https://demo-argo.itlinux.cc       # HTTP/2 200
```

The whole point: after the one-time bootstrap, *adding the repo = ArgoCD pulls it and does the magic*. A push to the `argo` branch is the only action needed to deploy or change anything — your CI/CD never calls the ArgoCD CLI.

DNS routing + Access (who can reach the hostname)

How DNS works with the tunnel

The public hostname must resolve to **the tunnel**, not to any server IP. `cloudflared tunnel route dns` creates a **proxied CNAME** in Cloudflare DNS:

```
demo-argo.itlinux.cc  CNAME  <tunnel-id>.cfargotunnel.com  (proxied / orange-cloud)
```

```
cloudflared tunnel route dns demo-argo demo-argo.itlinux.cc
dig +short demo-argo.itlinux.cc      # returns Cloudflare proxy IPs, NOT the cfargotunnel target
```

- The hostname here **must match** the `ingress:` hostname in the ConfigMap.
- The record is **proxied** (orange cloud) — that's what routes eyeball traffic through Cloudflare into the tunnel. A grey-cloud/DNS-only record will NOT work.
- One tunnel can serve many hostnames — add more `ingress:` entries and a `route dns` per hostname.

Option A — Open to everyone (public service)

Just the tunnel + DNS above. Anyone on the internet can reach `https://demo-argo.itlinux.cc`; the connector proxies to the in-cluster Service. Use for public sites/APIs. No identity check.

Option B — Gated by Cloudflare Access (authenticated users only)

Put **Zero Trust Access** in front of the same hostname so only authorized identities (Entra/Okta/One-time PIN, etc.) can reach it. The tunnel/DNS/ArgoCD setup is unchanged — Access is a policy layer on the hostname.

Dashboard: **Zero Trust** → **Access** → **Applications** → **Add** → **Self-hosted** - **Application domain:** `demo-argo.itlinux.cc` - **Identity providers:** select your IdP(s) (e.g. Entra) and/or One-time PIN - **Policy:** Allow → Include → emails / email-domain / IdP groups - Save. Now hitting the hostname shows the Access login page first; only users who pass the policy reach the cloudflared origin.

Terraform equivalent (self-hosted Access app + policy):

```
resource "cloudflare_zero_trust_access_application" "demo_argo" {
  account_id      = var.cloudflare_account_id
  name            = "Demo Argo (k8s)"
  domain         = "demo-argo.itlinux.cc"
  session_duration = "1h"
  # Show the login picker (SSO + One-time PIN); no auto-redirect.
  allowed_idps    = compact([var.entra_idp_id, var.otp_idp_id])
  auto_redirect_to_identity = false
}

resource "cloudflare_zero_trust_access_policy" "demo_argo_allow" {
  application_id = cloudflare_zero_trust_access_application.demo_argo.id
  account_id    = var.cloudflare_account_id
  name          = "Allow team"
  decision      = "allow"
  precedence    = 1
  include       = [{ email_domain = { domain = "itlinux.cc" } }]
}
```

	Open (Option A)	Access-gated (Option B)
Who reaches it	anyone	only identities passing the policy
Login screen	none	Cloudflare Access (SSO / PIN)
Extra setup	tunnel + DNS only	+ Access application & policy
Tunnel/ArgoCD config	same	same (Access is hostname-layer)

Access protects the **hostname at the edge** — it does not change the tunnel, the ConfigMap, or the ArgoCD sync. You can flip a hostname between open and gated without touching the cluster.

Terraform (optional) — manage DNS + Access as code

The `terraform/` directory provides a **hybrid IaC** option: the `demo-argo` tunnel/connector stays **manual** (CLI or dashboard, Step 1), and Terraform manages the two Cloudflare-side pieces that benefit from code — the **DNS route** and the **Access auth gate**. The cluster/connector side stays GitOps (ArgoCD).

```
terraform/
providers.tf          # cloudflare provider (~> 5.0)
variables.tf          # inputs (token, account, zone, tunnel_id, access toggle)
dns.tf               # proxied CNAME hostname -> <tunnel_id>.cfargotunnel.com
access.tf            # self-hosted Access app + allow policy (count on access_enabled)
outputs.tf           # hostname URL, CNAME target, access status
terraform.tfvars.example # copy to terraform.tfvars (gitignored) and fill
```

Usage:

```
cd terraform
cp terraform.tfvars.example terraform.tfvars # fill token, account, zone, tunnel_id
terraform init
terraform apply
```

- `access_enabled = true` → creates the Access app + policy (SSO/PIN gate).
- `access_enabled = false` → only the DNS route is managed; hostname is open.
- `tunnel_id` is the **manually-built** tunnel's id (`cloudflared tunnel list`).

This replaces the manual `cloudflared tunnel route dns` (DNS) and the dashboard Access steps with `terraform apply`, while leaving tunnel creation + the k8s connector (ArgoCD) exactly as documented above.

How redeployments work (GitOps)

With `automated` + `selfHeal` + `prune`, ArgoCD continuously reconciles the cluster to the `argo` branch:

You do...	ArgoCD does...
Push a manifest change (image tag, ingress, replicas)	Auto-syncs it to the cluster
<code>kubectl edit</code> manual drift	Reverts it back to git state (selfHeal)
Remove a resource from git	Prunes it from the cluster
(anything to <code>cloudflared-credentials</code>)	Ignores it — never managed/pruned

Redeploy = `git push` to `argo`. No manual `kubectl apply` needed.

Troubleshooting

- **Tunnel flaps** (`control stream encountered a failure`, `datagram handler ... context canceled`): QUIC datagrams hitting a reduced MTU on an overlay CNI (Calico VXLAN/IPIP, EKS). Already mitigated by `protocol: http2` in the ConfigMap. After editing, ArgoCD re-syncs; or `kubectl -n cloudflared rollout restart deploy/cloudflared`.

- **no such host / 502 for the origin:** the backend Service named in `ingress` doesn't exist in that namespace. Create it (e.g. `kubecttl create deployment front --image=nginx && kubecttl expose deployment front --port=80`).
- **Hostname won't resolve:** ensure the DNS route exists — `cloudflared tunnel route dns demo-argo demo-argo.itlinux.cc`.